PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	Chronic high altitude exposure and the epidemiology of ischemic stroke: A systematic review
AUTHORS	Ortiz-Prado, Esteban; Cordovez, Simone; Vasconez, Eduardo; Viscor, Ginés; Roderick, Paul

VERSION 1 – REVIEW

REVIEWER	Vecchi, Simona DEP Lazio
REVIEW RETURNED	28-Jun-2021

GENERAL COMMENTS	This paper aims to provide an overview of the literature on impact of high altitude exposure and risk of ischemic stroke. To explore this association the authors performed a systematic review. Systematic approach includes some methodological steps so I would request that the authors give more details on the method, data synthesis and results. Overall, the content of this review article is not clearly structured. Several major revision comments are listed in a separate file High altitude exposure and the epidemiology of ischemic stroke, A literature review This paper aims to provide an overview of the literature on impact of high altitude exposure and risk of ischemic stroke. To explore this association the authors performed a systematic review. Systematic approach includes some methodological steps so I would request that the authors give more details on the method, data synthesis and results. Overall, the content of this review article is not clearly structured. Several major revision comments are listed as below: Line 49-54: Study design. The sentence have too much redundant details. “To answer the link between hypobaric hypoxia due to high altitude exposure and stroke we performed a a literature review of the available information was performed. The first step prior the analysis was conducting a review of the existing published evidence. To conduct a comprehensive study of the available literature we performed a systematic search within the main bibliographic databases (see below). The review search was conducted….”

	Page 8. I would rename “Information sources” in Data sources and search strategy Page 8. line 8. Please, define date of the searches for each database. Page 8. Line 35. Inclusion and exclusion criteria List the inclusion criteria Define the inclusion criteria following PICO approach, then exclusion criteria. What population? (general population, only adults, only elderly people?) What type of study design? (all observational studies, only prospective observational studies, only studies with a comparator group) Exposure= what do you mean for high altitude? Over 1500 metres? What are the outcomes of interest? We considered only study with a cut-off point of short or long-term exposure in 28 days based on previously published data . List the exclusion criteria This literature review We excluded all in vivo and in vitro studies, and all the data analysed that did not concern human studies. If the search term "stroke" referred to cardiac output or stroke volume, the results were also excluded. Case reports were excluded from the study. The data extraction length was set from 1960 to 2019, and any study that matched the search strategy and did not counterpose the exclusion criteria was revised . We considered only study with a cut-off point of short or long-term exposure in 28 days based on previously published data. Page 8. Line 46. Bias assessment Evaluation of risk of bias is a critical step of a systematic review. Please give some detail about the CASP tool. CASP has appraisal checklists designed for different study design. Please describe the checklist you used and how this information is to be used in any data synthesis. General comment. In the method section you should describe data synthesis. Please add this paragraph. Results Page 9. Line 26. How many references you find? eleven? In this case, please check the number of references in the brackets. General comment. PRIMA guidelines require details on excluded studies and reason for exclusion. You excluded 167 articles. Please give a general overview of the reason of exclusion. Prisma flow chart diagram. Check and delete the box “included in quantitative synthesis”. You did non perform any quantitative synthesis. Page 9. Line 46. Specific studies Please give a general narrative description and synthesis of the included studies, study design, country, number of people involved (mean value), characteristics of the studies (gender, age,

	comorbidity). Perhaps you can describe the results on the base of study design. In the results you should synthetize and discuss the results of your assessment of quality of the included studies. Page 10. Line 9. You have included in your review Garcia J et al. 1994 reporting that “the study focused on the association between risk factors and stroke but not in the relationship with elevation”. Why do you include the study if it doesn’t meet inclusion criteria? General comment. I would request that the authors consider the following studies: 1.Lu et al. 2020. Characteristics of acute ischemic stroke in hospitalized patients in Tibet: a retrospective comparative study. BMC Neurol. 2020; 20: 380. 2. Sudhir Kumar Jha 1, Anil C Anand, Vivek Sharma, Nikhil Kumar, Chandra M Adya. Stroke at high altitude: Indian experience. High Alt Med Biol 2002;3(1):21-7.

REVIEWER	Minozzi, Silvia Lazio Regional Health Service, Rome
REVIEW RETURNED	30-Jul-2021

GENERAL COMMENTS	The aim of the authors, according to the abstract, is to perform a systematic review of the effect of acute or chronic exposure to hypoxia due to high-altitude on risk of stroke. Although the topic covered by the review could be of interested, the review suffers of critical methodological shortcomings and a very poor reporting that preclude the publication on a relevant journal such as BMJ open. Particularly, there is inconsistency between the objective reported in the abstract and the objective and inclusion criteria reported in the main text of the review; Results reported in the abstract and the conclusion of the review refer to studies on acute exposure, that were not included in the review. It appears clear, from several sentences, that the authors do not have the adequate expertise and methodological competences to conduct a systematic review Abstract Methods: The date of the search should be provided. The inclusion criteria for types of studies should also be provided Results: in the sentence reporting the results of the study selection process, authors use the terms “included” and “meeting the inclusion criteria” with different meanings, generating confusion about the actual number of the studies finally included. I suggest to rephrase the sentence “The authors included 191 manuscripts, while the rest of the abstracts were removed from the process for not meeting the inclusion criteria” as: “ The authors acquired in full text as potentially relevant 191 studies. Only 11 documents met the inclusion criteria and were finally included…. Conclusions: the sentences reported in the conclusions are actually the results of the review and should be reported in the results section of the abstracts. The authors stated in the objective that the review was aimed at assessing the irk of stroke due to acute or chronic exposure to high altitude, but in the results this distinction is not mentioned clearly: it seems that the risk of exposure at lower altitudes refer to chronic exposure while the risk

	of exposure to higher altitude refer to acute exposure; but nothing is reported about risk of choric exposure to higher altitude and acute exposure to lower altitude. Strengths and limitation of this studies La last sentence (point 4) of this section is incorrect: the state that "This review only found observational studies, therefore causality cannot be inferred from this analysis before completing further research." This is not true, as well conducted cohort studies are the best (and unique) source of evidence to make causal inference between and exposure and an outcome; RCTs on exposure to potentially risk factor are unethical and unfeasible, On the other hand, observational studies are very different between themselves, ranging from well conducted and large prospective cohort studies to very small and biased case series or cross sectional studies. Not all the observational studies can provide valid conclusions about causality. The authors should reformulate the sentences describing which kind of observational studies were included to support their conclusions that causality cannot be inferred Point 2 and 3 state more or less the same things; they could be combined in a unique point Objective section: the statement reported in the full review contradicts the statement reported in the objective section of the abstract: while in the abstract it is stated that " This review aims to provide an overview of the available evidence on risk prevalence, risk of stroke incidence, and or mortality when exposed to acute or chronic hypoxia" , in the full manuscript, objective section is reported that "we have conducted a literature review of the available information in terms of the link between high-altitude chronic exposure and the risk of developing ischemic stroke." . Moreover, the research question has been reported as" Does living at high-altitude increases the risk of developing ischemic stroke? " The effect of the acute exposure is no mentioned at all. Study design: the sentence "The review search was conducted in accordance with preferred reporting for systematic reviews and meta-analyses (PRISMA) guidelines." Is incorrect as the PRISMA concerns with the adequate and complete reporting of the entire review, not just the methods to conduct the bibliographic search. The statement " No protocol was registered in PROSPERO for this review because of its small size." Is unclear to me; why a systematic review that is expected to retrieve a small number of studies should not be registered in PROSPERO? Search strategies: here is reported that the inclusion criteria dealt only with chronic high-altitude exposure. Therefore, the abstract (methods and results) are not consistent with the review objective and methods Bias assessment: is not clear to me what the authors mean with term "confirmation bias" while referring to CASP tool Results. Although the authors stated that risk of bias /methodological quality was assessed by the CASP tools, they do not report the results of the assessment Discussion. The effect of acute exposure to high altitude is discussed only in the discussion section, citing studies that have not been listed among the included studies. While this could be appropriate to enlarge the discussion to topics not directly covered by the review objective , the results of such studies should not be reported in the abstract

	Conclusions: Any conclusions about of the effect of acute exposure should not be made, as the objective of the review was to assess the effect chronic exposure to high altitude

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1

Dr. Simona Vecchi, DEP Lazio

Comments to the Author:

This paper aims to provide an overview of the literature on impact of high altitude exposure and risk of ischemic stroke. To explore this association the authors performed a systematic review. Systematic approach includes some methodological steps so I would request that the authors give more details on the method, data synthesis and results. Overall, the content of this review article is not clearly structured. Several major revision comments are listed in a separate file

Thanks so much for your observations, we have resolved all your queries and accepted all your suggestions

Line 49-54: Study design. The sentence have too much redundant details.

“To answer the link between hypobaric hypoxia due to high altitude exposure and stroke we performed a a literature review of the available information was performed. The first step prior the analysis was conducting a review of the existing published evidence. To conduct a comprehensive study of the available literature we performed a systematic search within the main bibliographic databases (see below). The review search was conducted....”

we have shortened the sentences as follow:

A systematic review was conducted in accordance with the preferred reporting guidelines for systematic reviews and meta-analyses (PRISMA). No protocol was registered for this review since PROSPERO does not currently accept registrations for scoping reviews, literature reviews or mapping reviews.

Page 8. I would rename “Information sources” in Data sources and search strategy

Done

Page 8. line 8. Please, define date of the searches for each database.

Done

Page 8. Line 35. Inclusion and exclusion criteria

List the inclusion criteria

Define the inclusion criteria following PICO approach, then exclusion criteria.

What population? (general population, only adults, only elderly people?)

What type of study design? (all observational studies, only prospective observational studies, only studies with a comparator group)

Exposure= what do you mean for high altitude? Over 1500 metres?

What are the outcomes of interest?

We considered only study with a cut-off point of short or long-term exposure in 28 days based on previously published data.

Thanks, we have included this within the inclusion criteria section

List the exclusion criteria

This literature review We excluded all in vivo and in vitro studies, and all the data analyzed that did not concern human studies. If the search term "stroke" referred to cardiac output or stroke volume, the results were also excluded. Case reports were excluded from the study. The data extraction length was set from 1960 to 2019, and any study that matched the search strategy and did not counterpose the exclusion criteria was revised.

We considered only study with a cut-off point of short or long-term exposure in 28 days based on previously published data.

Thanks, we have clarified this in a separate section called “exclusion criteria”

Page 8. Line 46. Bias assessment

Evaluation of risk of bias is a critical step of a systematic review. Please give some detail about the CASP tool. CASP has appraisal checklists designed for different study design. Please describe the checklist you used and how this information is to be used in any data synthesis.

We have expanded in this using a table with every individual assessment which was placed within the supplementary file

General comment. In the method section you should describe data synthesis. Please add this paragraph.

Done

Results

Page 9. Line 26. How many references you find? eleven? In this case, please check the number of references in the brackets.

We have check the references and cite them all in this section

General comment. PRIMA guidelines require details on excluded studies and reason for exclusion. You excluded 167 articles. Please give a general overview of the reason of exclusion.

We have included a detailed description of why we excluded those studies

Prisma flow chart diagram. Check and delete the box “included in quantitative synthesis”. You did non perform any quantitative synthesis.

Thanks for pointing this out, we have amended the chart

Page 9. Line 46. Specific studies

Please give a general narrative description and synthesis of the included studies, study design, country, number of people involved (mean value), characteristics of the studies (gender, age, comorbidity). Perhaps you can describe the results on the base of study design.
In the results you should synthetize and discuss the results of your assessment of quality of the included studies.

Great suggestion, we have done so

Page 10. Line 9. You have included in your review Garcia J et al. 1994 reporting that "the study focused on the association between risk factors and stroke but not in the relationship with elevation".

Why do you include the study if it doesn't meet inclusion criteria?

In this study, the author discusses cerebrovascular events in people living above 2,500 m but does not analyze the role of hypoxia as a triggering or protective factor. We came to this work because it was conducted at high altitudes and had stroke as an outcome, nevertheless, no coments about hypoxia was given in this study.

General comment. I would request that the authors consider the following studies:

1.Lu et al. 2020. Characteristics of acute ischemic stroke in hospitalized patients in Tibet: a retrospective comparative study. BMC Neurol. 2020; 20: 380.

2. Sudhir Kumar Jha 1, Anil C Anand, Vivek Sharma, Nikhil Kumar, Chandra M Adya. Stroke at high altitude: Indian experience. High Alt Med Biol 2002;3(1):21-7

Thanks so much, we have included the one by Sudhir Kumar Jha et al previously, now we have included your suggested references as well as other that we found in a further search

Reviewer: 2

Dr. Silvia Minozzi, Lazio Regional Health Service, Rome

Comments to the Author:

The aim of the authors, according to the abstract, is to perform a systematic review of the effect of acute or chronic exposure to hypoxia due to high-altitude on risk of stroke. Although the topic covered by the review could be of interested, the review suffers of critical methodological shortcomings and a very poor reporting that preclude the publication on a relevant journal such as BMJ open. Particularly, there is inconsistency between the objective reported in the abstract and the objective and inclusion criteria reported in the main text of the review; Results reported in the abstract and the conclusion of the review refer to studies on acute exposure, that were not included in the review. It appears clear, from several sentences, that the authors do not have the adequate expertise and methodological competences to conduct a systematic review

Thanks for your comments and observations. We have improved the writing and the methodology; thus, we hope this revised version is clearer.

Abstract

Methods: The date of the search should be provided. The inclusion criteria for types of studies should also be provided

We have included this information

Results: in the sentence reporting the results of the study selection process, authors use the terms “included” and “meeting the inclusion criteria” with different meanings, generating confusion about the actual number of the studies finally included. I suggest to rephrase the sentence “The authors included 191 manuscripts, while the rest of the abstracts were removed from the process for not meeting the inclusion criteria” as: “ The authors acquired in full text as potentially relevant 191 studies. Only 11 documents met the inclusion criteria and were finally included….

Thanks for your important observations, we have added all your suggestions and we have summarized the results clearer.

Conclusions: the sentences reported in the conclusions are actually the results of the review and should be reported in the results section of the abstracts. The authors stated in the objective that the review was aimed at assessing the irk of stroke due to acute or chronic exposure to high altitude, but in the results this distinction is not mentioned clearly: it seems that the risk of exposure at lower altitudes refer to chronic exposure while the risk of exposure to higher altitude refer to acute exposure; but nothing is reported about risk of choric exposure to higher altitude and acute exposure to lower altitude.

Thanks so much for your observations, we have rephrased the entire abstract and reviewed the entire manuscript as well. A more structures and understandable abstract was created

Strengths and limitation of this studies

La last sentence (point 4) of this section is incorrect: the state that “This review only found observational studies, therefore causality cannot be inferred from this analysis before completing further research.” This is not true, as well conducted cohort studies are the best (and unique) source of evidence to make causal inference between and exposure and an outcome; RCTs on exposure to potentially risk factor are unethical and unfeasible, On the other hand, observational studies are very different between themselves, ranging from well conducted and large prospective cohort studies to very small and biased case series or cross sectional studies. Not all the observational studies can provide valid conclusions about causality. The authors should reformulate the sentences describing which kind of observational studies were included to support their conclusions that causality cannot be inferred

Point 2 and 3 state more or less the same things; they could be combined in a unique point

Thanks so much for your suggestions and comments. We have redone the entire strengths and limitations section

Objective section: the statement reported in the full review contradicts the statement reported in the objective section of the abstract: while in the abstract it is stated that “ This review aims to provide an overview of the available evidence on risk prevalence, risk of stroke incidence, and or mortality when exposed to acute or chronic hypoxia” , in the full manuscript, objective section is reported that “we have conducted a literature review of the available information in terms of the link between high-altitude chronic exposure and the risk of developing ischemic stroke.” . Moreover, the research question has been reported as” Does living at high-altitude increases the risk of developing ischemic stroke? “ The effect of the acute exposure is no mentioned at all.

We have clarified the text within the manuscript, and we have outlined the objectives in a better way. We have also explained in more detail that acute hypoxia was not included since several case-reports have mentioned the role of acute exposure on the development of stroke and we preferred to only focus on chronic hypoxia among high altitude dwellers rather than high altitude visitors.

Study design: the sentence “The review search was conducted in accordance with preferred reporting for systematic reviews and meta-analyses (PRISMA) guidelines.” Is incorrect as the PRISMA concerns with the adequate and complete reporting of the entire review, not just the methods to conduct the bibliographic search.

Thanks for pointing this out, you are correct, and we have changed this sentence accordingly

The statement “ No protocol was registered in PROSPERO for this review because of its small size.” Is unclear to me; why a systematic review that is expected to retrieve a small number of studies should not be registered in PROSPERO?

Thanks for your observation, we have contacted PROSPERO and they are not currently accepting registrations for scoping reviews, literature reviews or mapping reviews. PROSPERO have sent us the following message: PROSPERO is therefore unable to accept your application or provide a registration number. This decision should not stop you from submitting your project for publication to a journal.

Search strategies: here is reported that the inclusion criteria dealt only with chronic high-altitude exposure. Therefore, the abstract (methods and results) are not consistent with the review objective and methods

We have reviewed the entire manuscript for inconsistencies and is now improved after your keen suggestions

Bias assessment: is not clear to me what the authors mean with term “confirmation bias” while referring to CASP tool

We have clarified this section and explaining that we used the CASP tool to assess the manuscripts in order to interpret them subjectively.

Results.

Although the authors stated that risk of bias /methodological quality was assessed by the CASP tools, they do not report the results of the assessment

We have included a new table using the CASP tool as a supplementary table

Discussion. The effect of acute exposure to high altitude is discussed only in the discussion section, citing studies that have not been listed among the included studies. While this could be appropriate to enlarge the discussion to topics not directly covered by the review objective , the results of such studies should not be reported in the abstract

Thanks for your observations, we have deleted that part from the abstract and avoided this involuntary error further again.

Conclusions: Any conclusions about of the effect of acute exposure should not be made, as the objective of the review was to assess the effect chronic exposure to high altitude

Thanks so much for your observation, we have re-formulate our conclusions.

VERSION 2 – REVIEW

REVIEWER	Minozzi, Silvia Lazio Regional Health Service, Rome
REVIEW RETURNED	18-Dec-2021

GENERAL COMMENTS	The quality of the paper was substantially improved. However, there are still minor revisions that should be done Abstract: Date of the search you wrote as date f the last search "2021". However, when the authors do not specify exact date (i.e., day and month of the last search) this is normally interpreted as the authors search up to the end of the year. In your case this is not possible, as you submitted the revised version the 14th December. Please provide the exact date, including day and month, of the last search. Types of included studies: what do you mean by "scientific studies"? please describe the study designs that you included as reported in the main text of the draft. Methods section: search strategy. Please provide the period covered by the search, with the exact date (day , month and year) of the last date Risk of bias: I still read the term "confirmation bias" which is not clear to me; please provide an explanation of what do you mean by this term or rephrase.

VERSION 2 – AUTHOR RESPONSE

Reviewer: 2

Dr. Silvia Minozzi, Lazio Regional Health Service, Rome

Comments to the Author:

The quality of the paper was substantially improved. However, there are still minor revisions that should be done

Thanks for your comments

Abstract: Date of the search you wrote as date f the last search “2021”. However, when the authors do not specify exact date (i.e., day and month of the last search) this is normally interpreted as the authors search up to the end of the year. In your case this is not possible, as you submitted the revised version the 14th December. Please provide the exact date, including day and month, of the last search.

Thanks so much for pointing this out, we have added the specific date of December 1st 2021 as the date we last retrieved information

Types of included studies: what do you mean by “scientific studies”? please describe the study designs that you included as reported in the main text of the draft.

Thanks, we have done so

Methods section:

search strategy. Please provide the period covered by the search, with the exact date (day , month and year) of the last date

Thanks, we have done so

Risk of bias: I still read the term “confirmation bias” which is not clear to me; please provide an explanation of what do you mean by this term or rephrase.

Thanks so much for this, we have clarified the term and explained within the sentence.

Reviewer: 2

Competing interests of Reviewer: I have no competing interest